

VLISSINGEN, Netherlands

WMO: 063100

Lat: 51.4414N

Lon: 3.5958E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	28.3	18.5	13.9	27.5	22.2	16.7	29.4	42.5	49.0	39.1	48.3	11.7	60	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	80.0	67.8	76.4	66.2	73.4	64.7	69.1	77.0	67.5	74.3	66.0	71.8	9.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.0	72.9	64.8	91.9	71.0	63.6	88.0	69.6	33.2	77.4	31.9	74.3	30.8	71.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.9	30.7	27.2	DB	23.1	87.6	4.9	4.1	19.6	90.6	16.7	93.0	14.0	95.3	10.4	98.3	(4)
(5)			WB	21.8	72.4	4.7	1.5	18.4	73.5	15.6	74.3	13.0	75.2	9.6	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	40.1	40.5	44.4	50.1	56.2	61.3	65.5	65.8	61.2	54.8	47.6	41.7	(6)
(7)		DBStd	10.59	6.15	5.63	4.94	5.43	5.44	4.71	4.82	4.21	4.02	4.83	4.96	6.08	(7)
(8)		HDD50	1202	309	265	180	63	7	0	0	0	0	15	101	262	(8)
(9)		HDD65	4731	773	685	638	448	277	132	51	38	127	316	523	723	(9)
(10)		CDD50	2115	1	1	7	65	201	339	480	489	337	164	28	3	(10)
(11)		CDD65	168	0	0	0	0	5	20	66	62	14	1	0	0	(11)
(12)		CDH74	599	0	0	0	1	16	82	271	206	23	0	0	0	(12)
(13)		CDH80	109	0	0	0	0	1	12	59	36	1	0	0	0	(13)
(14)	Wind	WSAvg	13.3	15.8	14.8	13.9	12.1	12.2	11.9	11.7	11.6	12.1	13.8	14.4	15.3	(14)
(15)	Precipitation	PrecAvg	29.3	2.2	1.8	2.0	1.7	2.0	2.5	2.6	2.9	2.8	3.0	3.1	2.9	(15)
(16)		PrecMax	38.1	5.7	4.0	4.0	3.2	3.9	4.9	6.5	8.0	8.3	9.3	5.7	6.3	(16)
(17)		PrecMin	16.9	0.1	0.1	0.4	0.3	0.2	0.2	0.5	0.3	0.2	0.3	0.7	0.6	(17)
(18)		PrecStd	4.9	1.1	1.0	1.0	0.8	1.0	1.3	1.3	1.8	1.7	1.8	1.2	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	53.1	60.5	71.0	75.9	81.4	85.9	84.3	77.3	68.4	60.3	54.8	(19)
(20)			MCWB	50.4	48.8	52.6	57.5	62.3	67.6	69.3	68.7	67.2	62.3	57.3	52.3	(20)
(21)		2%	DB	50.8	51.0	55.6	65.6	72.0	75.8	80.4	79.3	72.4	64.7	57.6	52.9	(21)
(22)			MCWB	48.7	48.0	49.7	55.9	61.4	65.2	67.6	68.4	64.8	60.5	55.0	50.8	(22)
(23)		5%	DB	48.9	49.2	52.6	61.1	68.0	71.7	76.5	75.5	69.5	62.8	55.9	51.3	(23)
(24)			MCWB	47.0	46.6	48.5	53.5	58.9	63.6	66.4	66.5	63.4	59.3	53.4	49.2	(24)
(25)		10%	DB	47.5	47.6	50.5	57.6	64.4	68.4	72.9	72.2	67.2	61.2	54.1	49.4	(25)
(26)			MCWB	45.6	45.5	47.4	51.7	57.5	61.5	65.0	65.0	62.0	58.1	51.6	47.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	50.3	53.0	59.3	65.8	69.0	71.3	71.3	68.2	62.9	57.6	52.9	(27)
(28)			MCD B	52.4	52.3	58.2	68.3	74.2	77.9	80.8	79.7	74.7	66.7	59.2	54.1	(28)
(29)		2%	WB	49.2	48.4	50.9	56.3	62.1	66.3	69.1	69.3	66.0	61.3	55.5	51.2	(29)
(30)			MCD B	50.5	50.3	54.2	63.8	68.8	73.7	77.4	76.8	70.8	64.2	57.4	52.6	(30)
(31)		5%	WB	47.3	46.9	49.3	54.1	59.9	64.2	67.4	67.5	64.4	60.0	53.9	49.4	(31)
(32)			MCD B	48.6	48.6	51.8	59.6	66.0	70.4	74.8	73.5	68.3	62.4	55.7	50.9	(32)
(33)		10%	WB	45.8	45.6	48.0	52.4	58.1	62.5	65.8	66.0	62.9	58.5	52.1	47.6	(33)
(34)			MCD B	47.2	47.3	50.3	56.9	63.5	67.4	71.8	71.3	66.5	60.8	53.8	49.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.6	8.2	10.2	10.6	10.4	10.0	8.9	7.6	6.1	5.9	(35)	
(36)			MCDBR	6.5	7.7	11.6	15.7	16.6	16.7	16.4	15.4	12.3	8.7	6.7	6.2	(36)
(37)			MCWBR	6.3	6.1	7.7	8.3	8.5	8.9	7.3	7.2	6.7	6.1	6.0	6.1	(37)
(38)		5% WB	MCDBR	6.4	7.0	10.0	14.1	14.7	14.5	14.5	13.8	10.8	8.2	6.5	6.3	(38)
(39)			MCWBR	6.5	6.0	7.3	8.0	8.3	8.5	7.3	7.2	6.7	6.2	6.2	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.355	0.390	0.415	0.424	0.420	0.424	0.415	0.393	0.371	0.348	0.330	(40)	
(41)		taud	2.369	2.352	2.259	2.225	2.250	2.279	2.277	2.319	2.370	2.409	2.381	2.345	(41)	
(42)		Ebn at Noon	212	241	253	260	264	266	262	258	250	232	205	192	(42)	
(43)		Edh at Noon	20	26	35	40	41	40	40	36	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	238	461	854	1400	1645	1774	1674	1432	1047	595	278	180	(44)	
(45)		RadStd	20	83	107	124	116	135	171	116	69	51	28	21	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-180	N/A	N/A	(46)
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page